

Due by: 12/02/2021

Student's Name: _____ Total points = 23

Problem 1: (3+3 = 6 points)

(a) Consider an ideal pn junction diode at $T = 300$ K operating in the forward-bias region. Calculate the change in diode voltage that will cause a factor of 10 increase in current. (b) Repeat part (a) for a factor of 100 increase in current.

Solution: Solution 8.1

In forward bias

$$I_f \cong I_s \exp\left(\frac{eV}{kT}\right)$$

Then

$$\frac{I_{f1}}{I_{f2}} = \frac{I_s \exp\left(\frac{eV_1}{kT}\right)}{I_s \exp\left(\frac{eV_2}{kT}\right)} = \exp\left[\left(\frac{e}{kT}\right)(V_1 - V_2)\right]$$

or

$$V_1 - V_2 = \left(\frac{kT}{e}\right) \ln\left(\frac{I_{f1}}{I_{f2}}\right)$$

(a)

$$\text{For } \frac{I_{f1}}{I_{f2}} = 10, \text{ then} \quad V_1 - V_2 = (0.0259) \ln(10)$$

$$\text{Or } V_1 - V_2 = 59.6 \text{ mV} \cong 60 \text{ mV}$$

(b)

$$\text{For } \frac{I_{f1}}{I_{f2}} = 100, \text{ then}$$

$$V_1 - V_2 = (0.0259) \ln(100)$$

$$\text{Or } V_1 - V_2 = 119.3 \text{ mV} \cong 120 \text{ mV}$$

Problem 2: (3+2+2+2 = 9 points)

Problem 8.8

A one-sided p⁺n silicon diode has doping concentrations of $N_a = 5 \times 10^{17} \text{ cm}^{-3}$ and $N_d = 8 \times 10^{15} \text{ cm}^{-3}$. The minority carrier lifetimes are $\tau_{n0} = 10^{-7} \text{ s}$ and $\tau_{p0} = 8 \times 10^{-8} \text{ s}$. The cross-sectional area is $A = 2 \times 10^{-4} \text{ cm}^2$. Calculate the (a) reverse-bias saturation current, and (b) the forward-bias current at (i) $V_a = 0.45 \text{ V}$, (ii) $V_a = 0.55 \text{ V}$, and (iii) $V_a = 0.65 \text{ V}$.

Solution:

For Silicon, $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$

$$(a) \quad J_s = en_i^2 \left[\frac{1}{N_a} \sqrt{\frac{D_n}{\tau_{n0}}} + \frac{1}{N_d} \sqrt{\frac{D_p}{\tau_{p0}}} \right] \quad D_n = \mu_n kT/q \text{ and } \mu_n \text{ is doping dependent}$$

see figure 3-23

$$D_p = \mu_p kT/q \text{ and } \mu_p \text{ is doping dependent see figure 3-23}$$

$$= 11 / \text{cm}^2$$

$$J_s = (1.6 \times 10^{-19}) (1.5 \times 10^{10})^2$$

$$\times \left[\frac{1}{5 \times 10^{17}} \sqrt{\frac{25}{10^{-7}}} + \frac{1}{8 \times 10^{15}} \sqrt{\frac{10}{8 \times 10^{-8}}} \right]$$

$$J_s = 5.145 \times 10^{-11} \text{ A/cm}^2$$

$$I_s = AJ_s = (2 \times 10^{-4}) (5.145 \times 10^{-11})$$

$$= 1.029 \times 10^{-14} \text{ A}$$

$$(b) \quad I = I_s \exp\left(\frac{V_a}{V_t}\right)$$

$$(i) \quad I = (1.029 \times 10^{-14}) \exp\left(\frac{0.45}{0.0259}\right) = 3.61 \times 10^{-7} \text{ A}$$

$$(ii) \quad I = (1.029 \times 10^{-14}) \exp\left(\frac{0.55}{0.0259}\right) = 1.72 \times 10^{-5} \text{ A}$$

$$(iii) \quad I = (1.029 \times 10^{-14}) \exp\left(\frac{0.65}{0.0259}\right) = 8.16 \times 10^{-4} \text{ A}$$

Problem 3: (2+2+2+2 = 8 points)

Fill in the missing data in the following table.

Case	V_a (V)	I (mA)	I_s (mA)	J_s (mA/cm ²)	A (cm ²)
1	0.65	0.50			2×10^{-4}
2	0.70		2×10^{-12}		1×10^{-3}
3		0.80		1×10^{-7}	1×10^{-4}
4	0.72	1.20		2×10^{-8}	

Solution **Case 1:** $I = I_s \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right]$ since exponent is (79.3×10^9) much much larger than 1 we can write

$$I = I_s \exp\left(\frac{V_a}{V_t}\right)$$

$$0.50 \times 10^{-3} = I_s \exp\left(\frac{0.65}{0.0259}\right) \Rightarrow I_s = 6.305 \times 10^{-15} \text{ A} = 6.305 \times 10^{-12} \text{ mA}$$

$$J_s = \frac{I_s}{A} = \frac{6.305 \times 10^{-12}}{2 \times 10^{-4}} = 3.153 \times 10^{-8} \text{ mA/cm}^2$$

Case 2: $I = I_s \exp\left(\frac{V_a}{V_t}\right)$

$$= (2 \times 10^{-12}) \exp\left(\frac{0.70}{0.0259}\right) \text{ or } I = 1.093 \text{ mA}$$

$$J_s = \frac{I_s}{A} = \frac{2 \times 10^{-12}}{1 \times 10^{-3}} = 2 \times 10^{-9} \text{ mA/cm}^2$$

Case 3: $I = AJ_s \exp\left(\frac{V_a}{V_t}\right)$ So $V_a = V_t \ln\left[\frac{I}{AJ_s}\right]$

$$= (0.0259) \ln\left[\frac{0.80}{(10^{-4})(10^{-7})}\right] \text{ or } V_a = 0.6502$$

Then $I_s = AJ_s = (10^{-4})(10^{-7}) = 10^{-11} \text{ mA}$

Case 4: $I_s = \frac{I}{\exp\left(\frac{V_a}{V_t}\right)} = \frac{1.20}{\exp\left(\frac{0.72}{0.0259}\right)} \text{ or } I_s = 1.014 \times 10^{-12} \text{ mA}$

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$$A = \frac{I_s}{J_s} = \frac{1.014 \times 10^{-12}}{2 \times 10^{-8}} = 5.07 \times 10^{-5} \text{ cm}^2$$